

DecisionIQ Master Handbook

CHAPTER 10: Master Data Management (MDM): Account Consolidation & Re-Keying

10.1 The Duplicate Customer Account Challenge

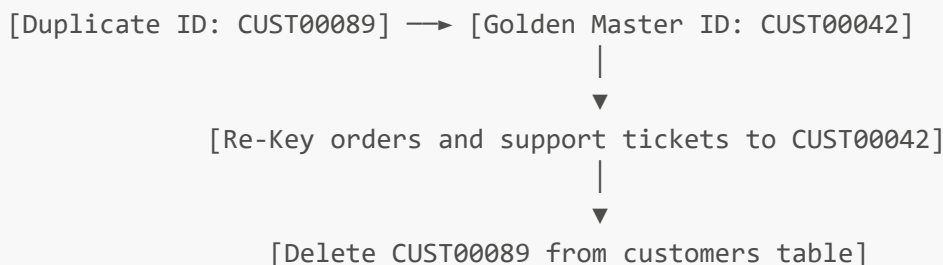
In enterprise databases, duplicate customer entries occur frequently. For example, a customer might register twice using different names or IDs but the same email address:

- CUST00042: John Doe | john.doe@company.com
- CUST00089: J. Doe | john.doe@company.com

If the ETL pipeline simply drops the duplicate customer record CUST00089 from the database, it will crash when trying to load transactions referencing the deleted customer ID, throwing a **Foreign Key constraint violation**.

10.2 Account Re-Keying Logic

To consolidate duplicate customer records without orphan transactions, the ETL pipeline builds a mapping dictionary and updates all associated records in child tables:



1. **Deduplicate Customers:** Group by email and keep the earliest record (Golden Master).
2. **Generate ID Mapping:** Extract a mapping dictionary linking the duplicate customer IDs to their master customer ID:

```
is_duplicate = customers_raw.duplicated(subset=["contact_email"],
keep="first")
master_ids = customers_raw[~is_duplicate][["contact_email",
"customer_id"]].rename(columns={"customer_id": "master_customer_id"})
duplicate_ids = customers_raw[is_duplicate][["contact_email", "customer_id"]]
mapping_df = duplicate_ids.merge(master_ids, on="contact_email", how="left")
customer_id_map = dict(zip(mapping_df["customer_id"],
mapping_df["master_customer_id"]))
```

3. **Map Child Records:** Update orders and customer_support customer IDs:

```
if customer_id_map:
    orders_clean["customer_id"] =
orders_clean["customer_id"].replace(customer_id_map)
    dfs["customer_support"]["customer_id"] = dfs["customer_support"]
["customer_id"].replace(customer_id_map)
```

This ensures database integrity remains intact during de-duplication.